KEYVAN
AVIATION

BAU Bahçeşehir University 25.yıl

CO-OP
KAMPÜSÜMİŞYERİM
KARİYER GELİŞTİRME UYGULAMA VE ARAŞTIRMA MERKEZİ

WEEK # 8

Meteorological & Operational Intelligence in Aviation

2026

Weather and NOTAM Data in Aviation

Weather is one of the **most critical factors affecting flight safety and efficiency**.

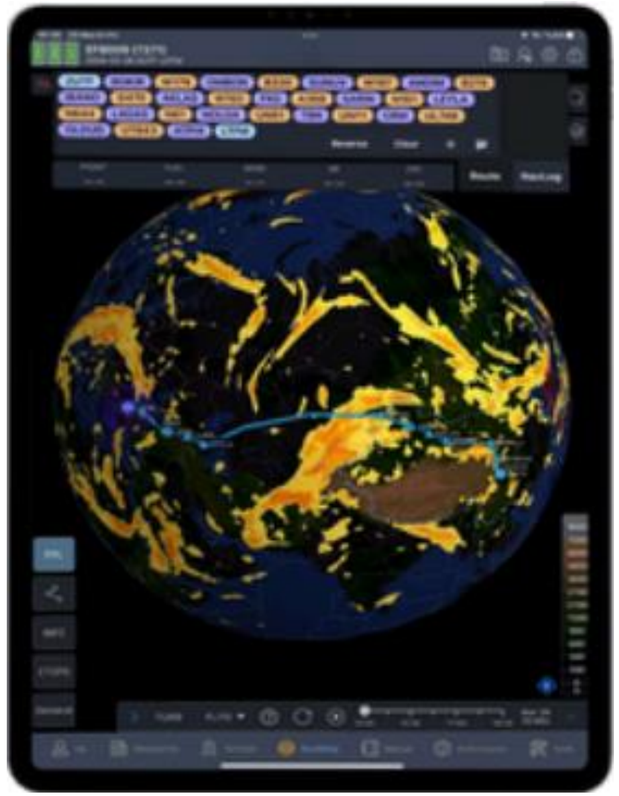
Pilots and dispatchers must constantly answer questions such as:

- Is it safe to depart?
- Can we land at the destination?
- Should we change the route?
- Do we need an alternate airport?

These decisions rely on **meteorological and operational data systems**.

Modern aviation safety depends on the **integration of three elements**:

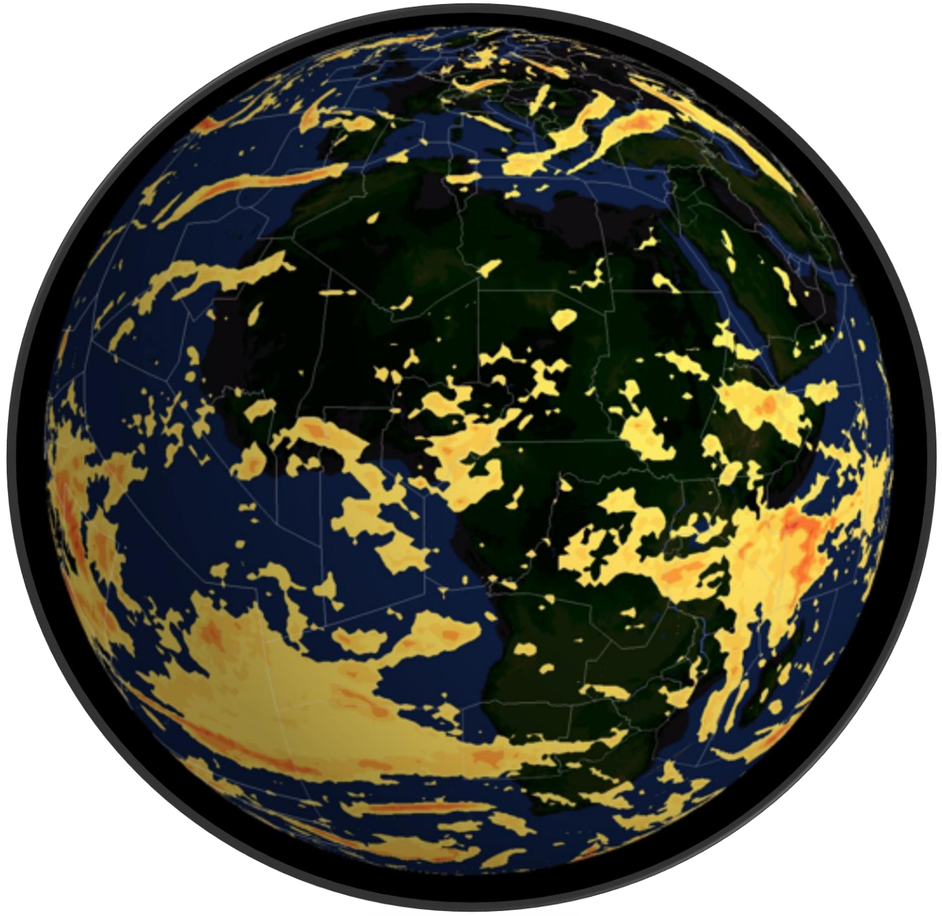
1. Meteorological observations
2. Aeronautical information (NOTAMs)
3. Digital charting systems



Keyvan Aviation © 2025. All rights reserved.

Meteorological Data (METAR, TAF, SIGMET etc.)

- Weather is one of the primary operational risk factors in aviation
- Most aviation weather hazards are invisible
- Weather influences all flight phases:
  - Pre-flight planning
  - Takeoff
  - En-route cruise
  - Approach & landing
- Weather uncertainty increases operational complexity
- Safe aviation depends on accurate interpretation of meteorological data



Keyvan Aviation © 2025. All rights reserved.

Meteorological Data (METAR, TAF, SIGMET etc.)

- **METAR** – Meteorological Aerodrome Report (actual observed weather)
- **TAF** – Terminal Aerodrome Forecast (predicted weather)
- **SIGMET** – Significant Meteorological Information (severe hazard warnings)
- **AIRMET** – Airmen’s Meteorological Information (moderate hazard advisories)
- **NOTAM** – Notice to Airmen (operational status of airspace or airport)
- **Satellite Imagery** – Cloud cover, convective systems, jet streams
- **Radar Data** – Precipitation, storm cells, turbulence, wind shear

Keyvan Aviation © 2025. All rights reserved.

Meteorological Data (METAR, TAF, SIGMET etc.)

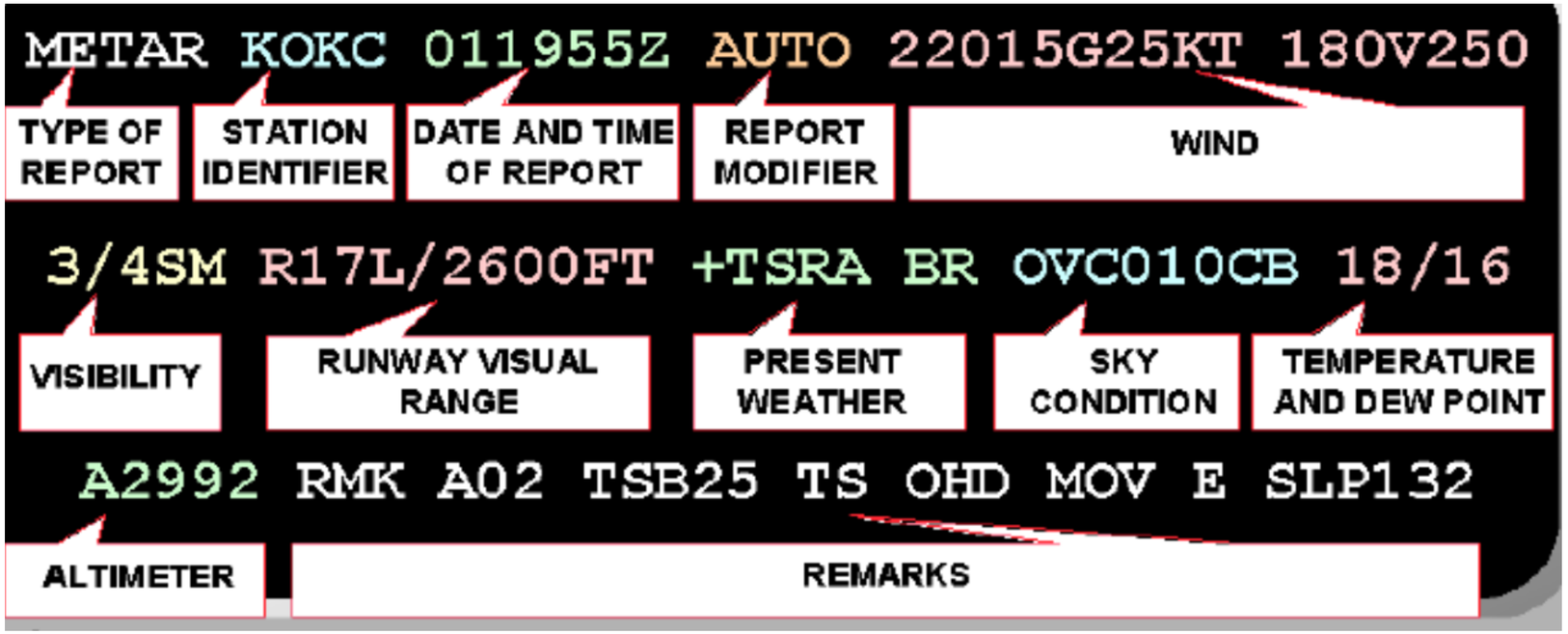
METAR – Current Weather Observation

METAR provides real-time weather conditions at an airport.
Typical information includes:

- Wind direction and speed
- Visibility
- Weather phenomena
- Cloud layers
- Temperature and dew point
- Atmospheric pressure

This information is essential for:

- Takeoff and landing safety
- Runway selection
- Visibility assessment



Keyvan Aviation © 2025. All rights reserved.

METAR – Current Weather Observation

LTFM 121250Z 02015KT 8000 -RA SCT020 BKN040 12/08 Q1015

Airport – Time – Wind – Visibility – Weather – Clouds – Temp/Dew – Pressure

•**LTFM** → Airport code (Istanbul Airport)
•**121250Z** → Day 12, time 12:50 UTC (Z = Zulu / UTC)
•**02015KT** → Wind from 020° at 15 knots
•**8000** → Visibility 8 kilometers
•**-RA** → Light rain (- = light, RA = rain)
•**SCT020** → Scattered clouds at 2000 ft
•**BKN040** → Broken clouds at 4000 ft
•**12/08** → Temperature 12°C, dew point 8°C (saturated %100 humidity)
•**Q1015** → Pressure at sea level 1015 hPa (altimeter setting, high pressure, low pressure etc)

Code	Meaning
RA	Rain
SN	Snow
TS	Thunderstorm
SCT	Scattered clouds
BKN	Broken clouds
OVC	Overcast
DZ	Drizzle
FG	Fog
BR	Mist

Keyvan Aviation © 2025. All rights reserved.

METAR – Current Weather Observation

•**Headwind (toward aircraft):**

•**Takeoff:** Aircraft generates lift sooner → **shorter takeoff roll** → better climb performance.
•**Landing:** Lower ground speed → **shorter landing distance** → improved control.
•Stronger headwind → **more pronounced benefits** for both takeoff and landing.

•**Tailwind (from behind):**
•**Takeoff:** Requires **longer runway** to reach lift-off speed → reduced performance.
•**Landing:** Higher ground speed → **longer landing distance** → higher overrun risk.
•Stronger tailwind → **greater runway needed**, often avoided if possible.

•**Crosswind (from side):**
•**Takeoff & Landing:** Aircraft may drift sideways → pilots must use **rudder** to maintain runway alignment.
•Stronger crosswind → **more difficult control**, may exceed aircraft limits.
Typical limits:
Large jets: **30–40 kt**
Medium aircraft: **20–30 kt**
Small aircraft: **10–20 kt**

For both takeoff and landing, pilots and ATC prefer **runways aligned with the wind** to maximize headwind and minimize crosswind, while avoiding tailwinds. Wind **strength** amplifies the effect of direction: strong headwind helps, strong tailwind or crosswind challenges control and safety.

METAR:
`02015KT`
Meaning:
•Wind **from 020°**
•Speed **15 knots**
If airport runways are:
•**02/20**
Best choice:
•**Runway 02 → 15 kt headwind**
•**Runway 20 → 15 kt tailwind**
Pilots choose **Runway 02**.

Keyvan Aviation © 2025. All rights reserved.

METAR – Current Weather Observation

Operational Importance of METAR

Runway selection: Wind direction and speed determine which runway to use

Landing minima: Visibility affects approach category requirements

Crosswind limitation: Aircraft have maximum crosswind limits

De-icing requirement: Temperature and moisture indicate possible ice formation

Takeoff performance: Pressure, temperature, and wind affect acceleration and takeoff distance

Keyvan Aviation © 2025. All rights reserved.

TAF – Terminal Aerodrome Forecast

Forecast for airport conditions: 24–30 hours forward
Includes: wind, visibility, weather phenomena, cloud trends, probability groups

TAF helps operators plan:

- Flight schedules
- Fuel requirements
- Alternate airports
- Possible weather deterioration

Keyvan Aviation © 2025. All rights reserved.

TAF – Terminal Aerodrome Forecast

TAF LTFM 271100Z 2712/2818	**Header Information**
02015KT 8000 BKN030	**Initial Forecast Conditions**
TEMPO 2714/2718 4000 -SHRA BKN015	**Temporary Change Group**
PROB30 2720/2723 TSRA	**Probability Forecast**
FM280000 03010KT CAVOK	**New Permanent Conditions**

Element	Meaning	Explanation
TAF	Terminal Aerodrome Forecast	Aviation forecast for an airport
LTFM	ICAO airport code	Istanbul Airport
271100Z	Issue time	Issued on the 27th at 11:00 UTC
2712/2818	Validity period	Valid from 27th 12:00 UTC until 28th 18:00 UTC

forecast covers **30 hours**

02015KT	Wind from 020° at 15 knots
8000	Visibility 8000 meters
BKN030	Broken clouds at 3000 ft

This describes the **expected prevailing conditions at the start of the forecast period**.

TEMPO	Temporary change
2714/2718	From 14:00 to 18:00 UTC
4000	Visibility may drop to 4 km
-SHRA	Light rain showers
BKN015	Broken cloud at 1500 ft

Weather may temporarily deteriorate between 14:00–18:00 UTC.

Keyvan Aviation © 2025. All rights reserved.

TAF – Terminal Aerodrome Forecast

TAF LTFM 271100Z 2712/2818	**Line 1 – Header Information**
02015KT 8000 BKN030	**Initial Forecast Conditions**
TEMPO 2714/2718 4000 -SHRA BKN015	**Temporary Change Group**
PROB30 2720/2723 TSRA	**Probability Forecast**
FM280000 03010KT CAVOK	**New Permanent Conditions**

PROB30	**30% probability**
2720/2723	**Between 20:00–23:00 UTC**
TSRA	**Thunderstorm with rain**

There is a **30% chance of thunderstorms**.

FM	From (abrupt permanent change)
280000	From the 28th at 00:00 UTC
03010KT	Wind 030° at 10 knots
CAVOK	Ceiling and visibility OK

After midnight conditions become **good and stable**.

Keyvan Aviation © 2025. All rights reserved.

SIGMET – Significant Meteorological Information

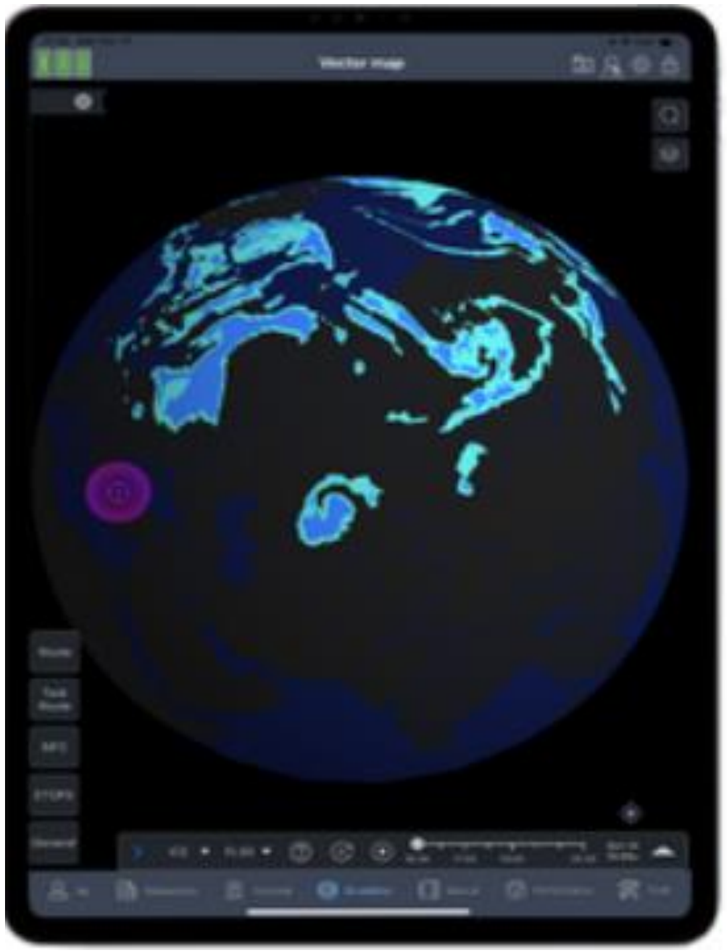
SIGMET warnings are issued for hazardous weather affecting flight safety.
Examples include:

- Severe turbulence
- Severe icing
- Volcanic ash
- Tropical cyclones
- Severe thunderstorms

SIGMETs apply to large regions such as FIRs.

SIGMET 3 VALID 271200/271600 LTAA-
LTAA ANKARA FIR SEV TURB OBS AT 1130Z
FL300/FL380 MOV NE 25KT
NC

SIGMET	Significant Meteorological Information
3	Message sequence number
VALID 271200/271600	Valid from 12:00 to 16:00 UTC on the 27th
LTAA	FIR concerned (Ankara FIR)



The warning applies to **aircraft flying in the Ankara Flight Information Region**.

Keyvan Aviation © 2025. All rights reserved.

SIGMET – Significant Meteorological Information

SIGMET 3 VALID 271200/271600 LTAA-	MESSAGE IDENTIFICATION
LTAA ANKARA FIR SEV TURB OBS AT 1130Z	Hazard Description
FL300/FL380 MOV NE 25KT	Vertical Extent/Movement
NC	Intensity Trend

LTAA	FIR identifier
SEV TURB	Severe turbulence
OBS	Observed
AT 1130Z	Observed at 11:30 UTC

Severe turbulence has been **observed in the Ankara FIR at 11:30 UTC**.

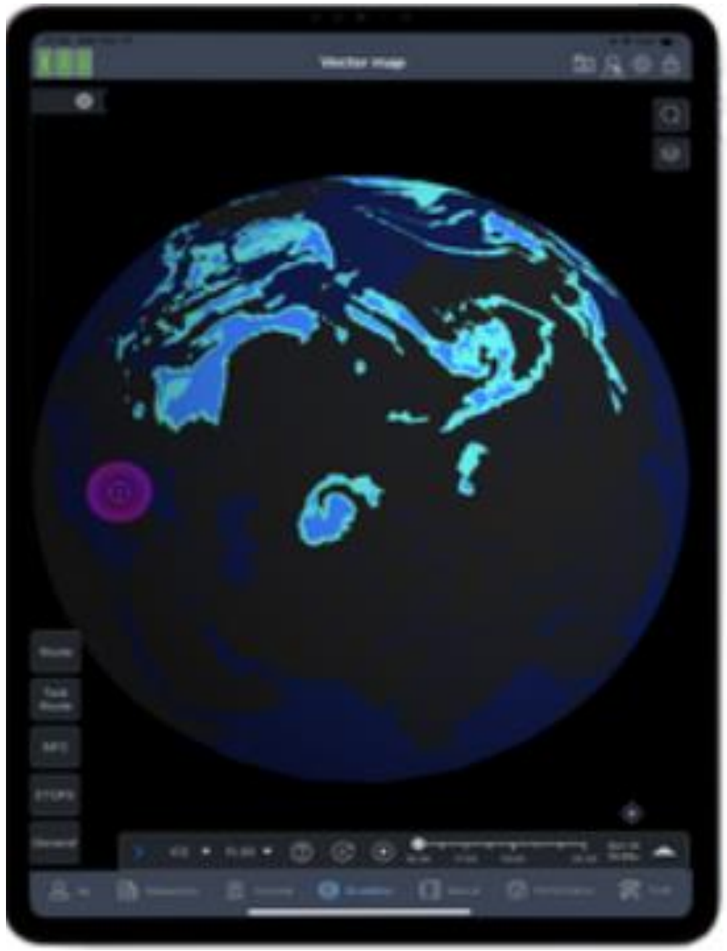
FL300	Flight Level 300 (30,000 ft)
FL380	Flight Level 380 (38,000 ft)

The turbulence occurs between **30,000 ft and 38,000 ft altitude**.

NE	Northeast
25KT	Speed 25 knots

The turbulence area is **moving northeast at 25 knots**.

NC	No change

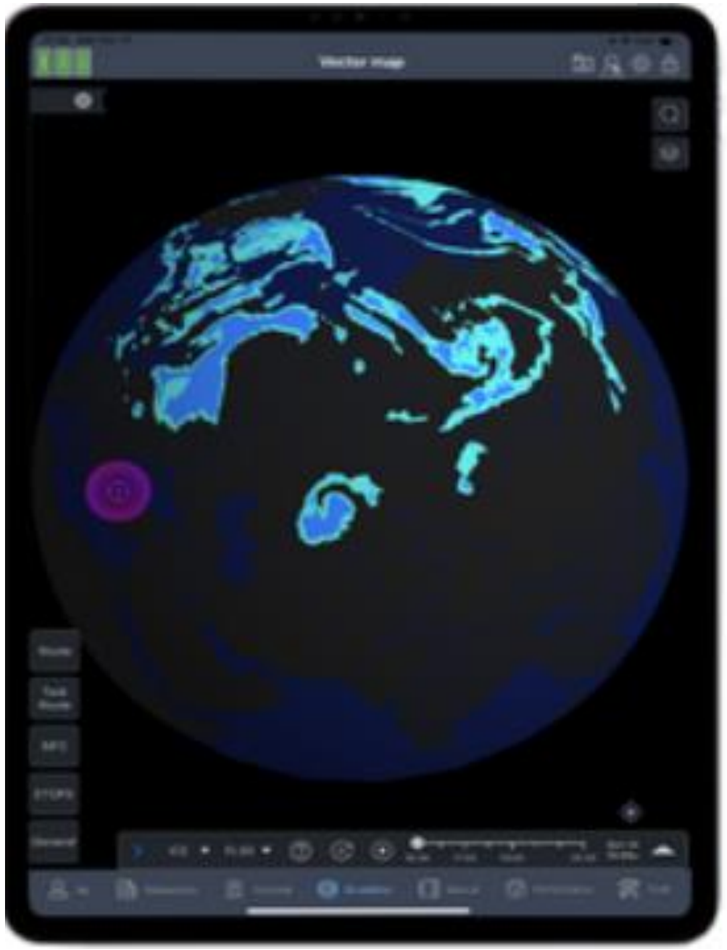


Keyvan Aviation © 2025. All rights reserved.

SIGMET – Significant Meteorological Information

Hazard	Abbreviation	
Severe turbulence	SEV TURB	Can be extreme; risk of structural damage or loss of control, Can cause injuries if passengers or crew are not secured
Severe icing	SEV ICE	Can reduce lift, increase drag, and cause instrument or engine malfunctions
Severe thunderstorms	TS	**Hail damage** to aircraft skin or windshields
Embedded thunderstorms	EMBD TS	
Squall line	SQL TS	
Tropical cyclone	TC	
Volcanic ash	VA	**Engines can fail** or be damaged by ash ingestio
Mountain waves	MTW	

INTSF	Intensifying
WKN	Weakening



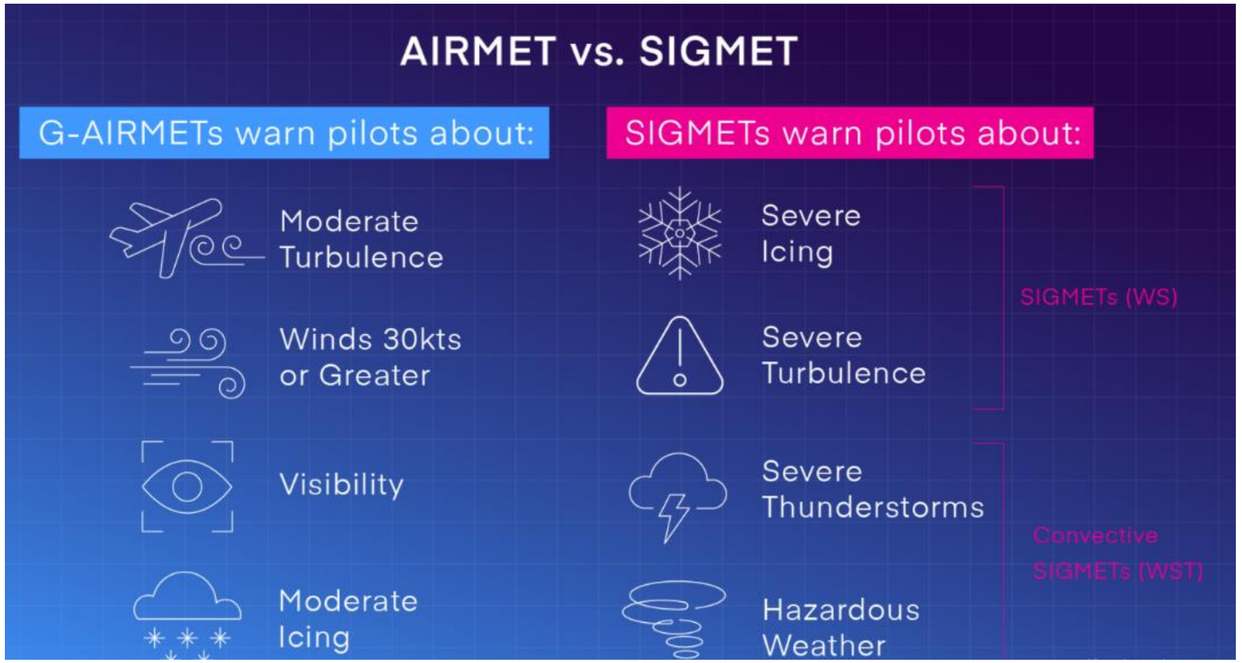
Keyvan Aviation © 2025. All rights reserved.

AIRMET (Airmen’s Meteorological Information)

AIRMET (Airmen’s Meteorological Information) is a meteorological advisory that provides information on weather phenomena of operational significance, which are less severe than SIGMET, but still important for flight safety, especially for low-level and general aviation operations.

The purpose of an AIRMET is to:
Warn pilots and ATC about widespread moderate weather phenomena
Support tactical decision-making, not emergency action
Improve situational awareness in a FIR or sub-FIR region

AIRMET = awareness and planning,
SIGMET = immediate action and avoidance



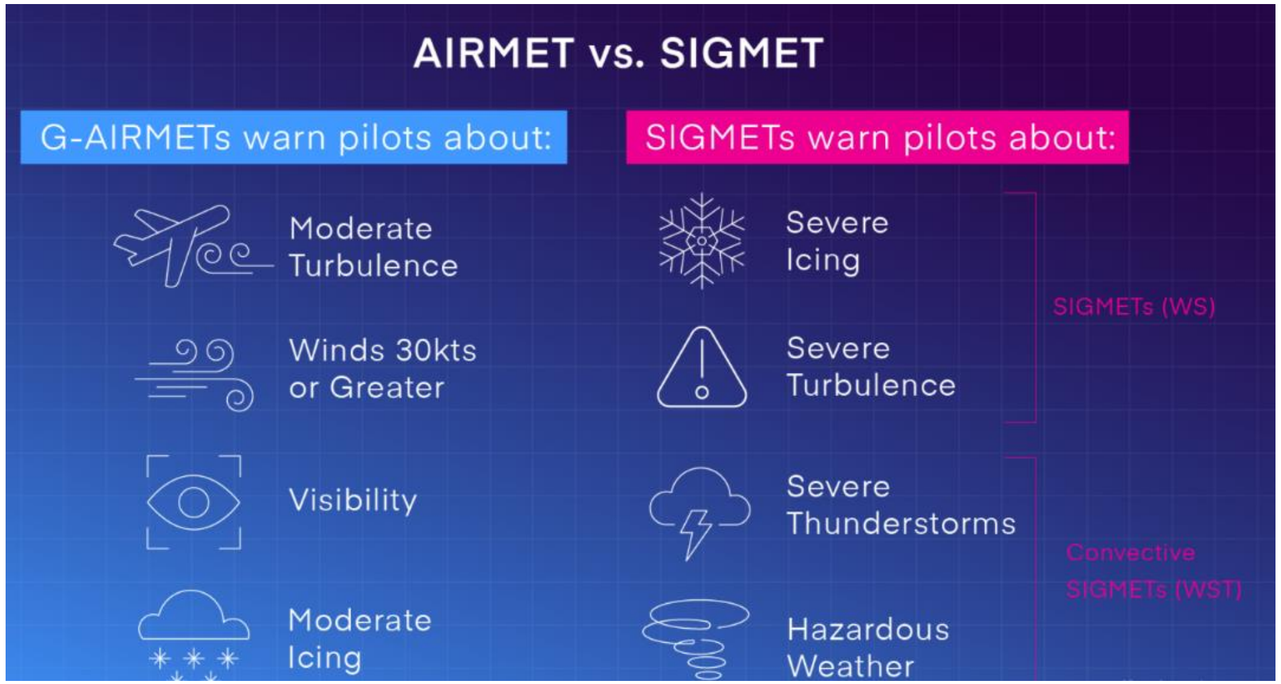
Keyvan Aviation © 2025. All rights reserved.

AIRMET (Airmen’s Meteorological Information)

AIRMET	Message Type
VALID 121200/121800	Validity Period
LTBB ISTANBUL FIR	Area of Application
MOD TURB BTN FL080 AND FL180	Hazard Description/Vertical Range
MOV NE 15KT	Movement of the Weather System

This AIRMET warns pilots about **moderate turbulence in the Istanbul Flight Information Region**.

LTBB	ICAO code for Istanbul FIR
FIR	Flight Information Region
MOD	Moderate
TURB	Turbulence
BTN	Between
FL080	Flight Level 80 (8,000 ft)
FL180	Flight Level 180 (18,000 ft)
MOV	Moving
NE	Northeast
15KT	Speed of 15 knots



For pilots and dispatchers, this AIRMET means:
Possible operational considerations:
•Expect turbulence during **climb or descent**
•Consider **alternative flight levels**
•Warn passengers to **keep seatbelts fastened**
•Cabin crew should secure cabin equipment
For example:
If a flight cruises at **FL350**, it will not experience turbulence at cruise altitude but **may encounter it during climb or descent**.

Keyvan Aviation © 2025. All rights reserved.

Meteorological Data (METAR, TAF, SIGMET etc.)

Feature	METAR	TAF	AIRMET	SIGMET
Type	Observation	Forecast	Advisory	Warning
Time reference	Current	Future	Current / Forecast	Current / Forecast
Geographic scope	Aerodrome	Aerodrome	FIR / Region	FIR / Defined Area
Vertical scope	Surface–low	Surface–low	Mainly low–mid	Mid–high
Severity	Descriptive	Predictive	**Moderate**	**Severe**
Main user	ATC, pilots	ATC, planners	Pilots, ATC	Pilots, ATC

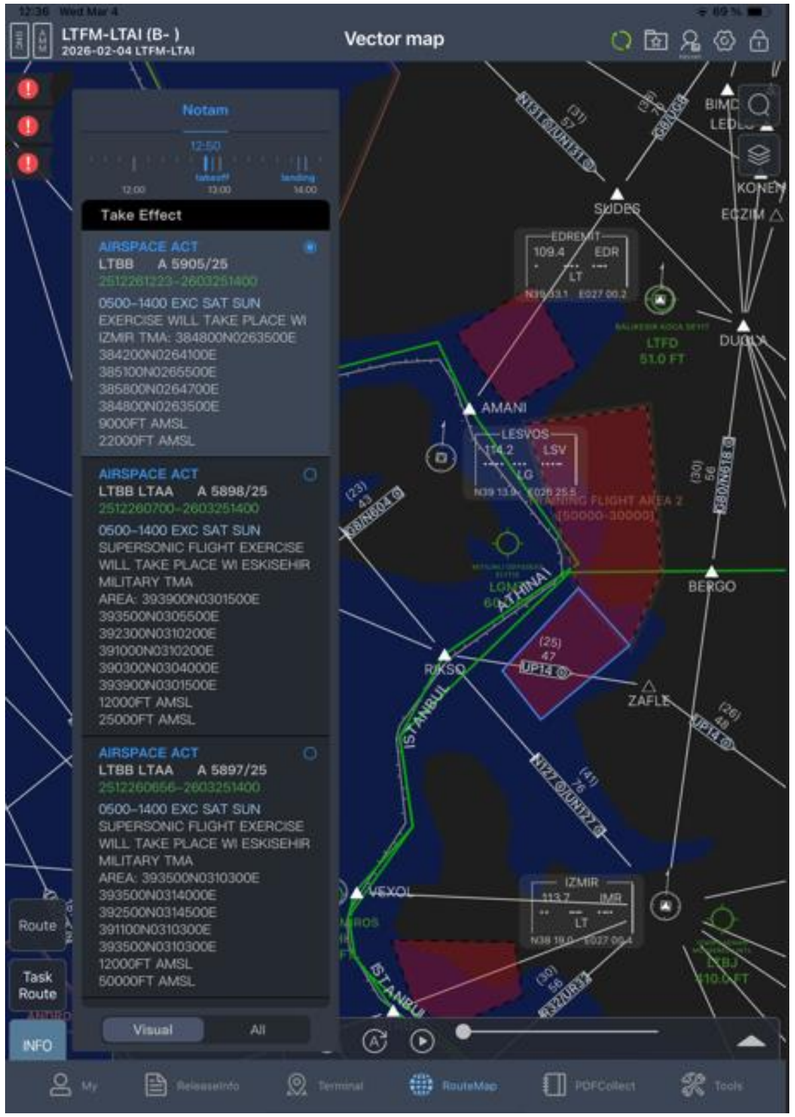
Keyvan Aviation © 2025. All rights reserved.

NOTAM

A **NOTAM (Notice to Air Missions)** is a short, time-critical message used to inform pilots and aviation stakeholders about **temporary or rapidly changing conditions** that affect flight operations but are **not yet published on charts or in the AIP**.

Charts show what is normally there. NOTAMs show what has changed.

NOTAMs are mandatory to check before every flight and are legally part of pre-flight briefing.



Why NOTAMs Exist
Aeronautical information is usually published on a **fixed cycle** (AIP, charts), but aviation operations change **daily or even hourly**. NOTAMs fill this gap by providing fast, standardized updates that cannot wait for the next publication cycle.

NOTAMs may include:
- Runway or taxiway closures
- Changes to approach or SID availability
- Navaid outages or limitations
- Obstacle status (cranes, masts)
- Airspace restrictions or activations
- Changes to lighting systems
- Temporary procedures

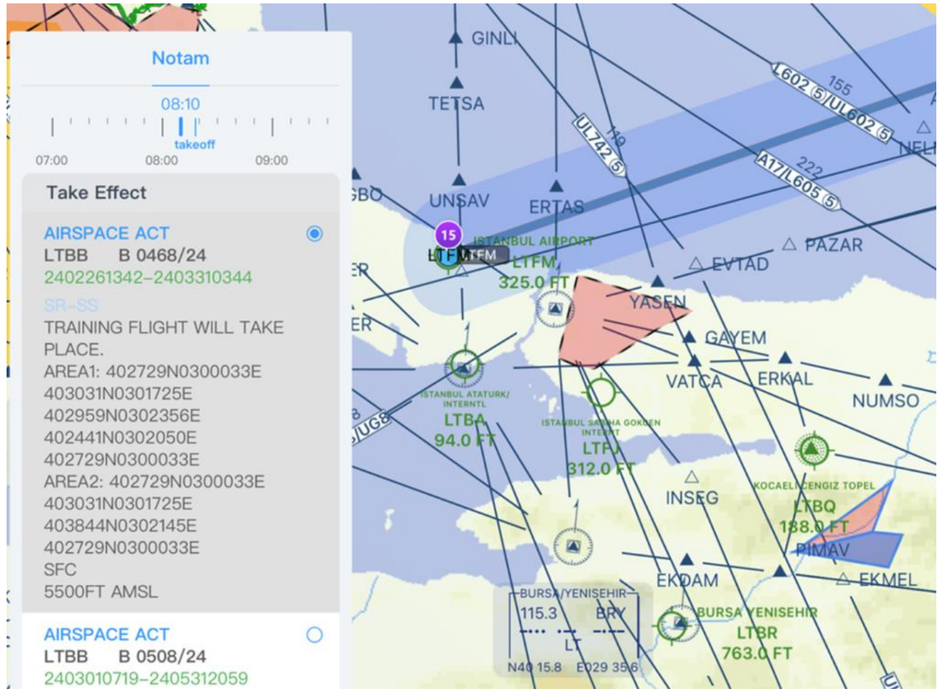
Keyvan Aviation © 2025. All rights reserved.

NOTAM

A1234/26 NOTAMN	Message Identification
Q) LTBB/QMRLC/IV/NBO/A/000/999/4100N02845E005	NOTAM Classification Line
A) LTFM	Aerodrome Identification
B) 2603151200	Start Time
C) 2603201800	End Time
E) RWY 16R/34L CLOSED DUE WIP	Operational Description

A1234	NOTAM serial number
/26	Year of issue (2026)
NOTAMN	New NOTAM
NOTAMR	Replacement of previous NOTAM
NOTAMC	Cancellation of previous NOTAM

LTBB	FIR identifier (Istanbul FIR)
QMRLC	Subject code (Runway closure)
IV	Applies to IFR and VFR traffic
NBO	Immediate attention for operators
A	Aerodrome-related NOTAM
000/999	Vertical limits (surface to unlimited)
4100N02845E	Coordinates of the location
005	Radius in nautical miles



This NOTAM concerns **runway closure at an aerodrome within Istanbul FIR**.

Keyvan Aviation © 2025. All rights reserved.

NOTAM

A1234/26 NOTAMN	Message Identification
Q) LTBB/QMRLC/IV/NBO/A/000/999/4100N02845E005	NOTAM Classification Line
A) LTFM	Aerodrome Identification
B) 2603151200	Start Time
C) 2603201800	End Time
E) RWY 16R/34L CLOSED DUE WIP	Operational Description

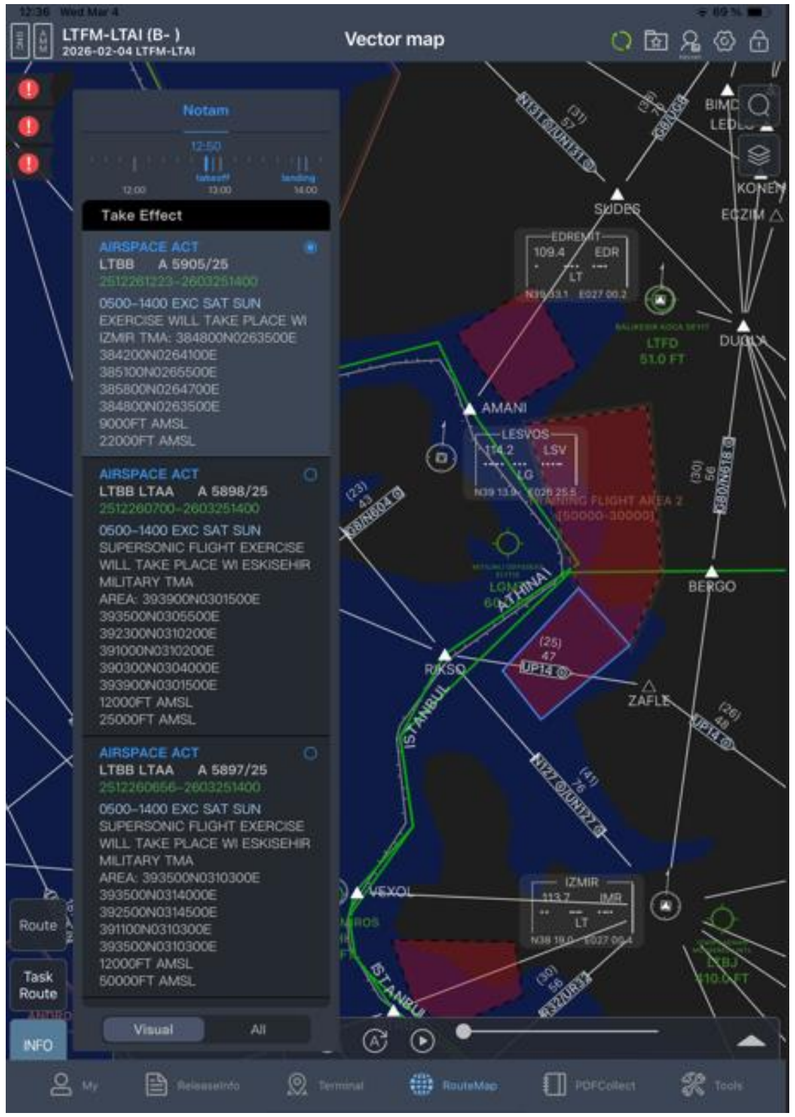
A)	Aerodrome location
LTFM	Istanbul Airport
26	Year (2026)
03	Month (March)
15	Day
1200	Time (UTC)

The NOTAM becomes active on: **15 March 2026 at 12:00 UTC**

The NOTAM remains valid until: **20 March 2026 at 18:00 UTC**
After that time the runway should be operational again.

RWY	Runway
16R/34L	Runway designations
CLOSED	Not available for operations
WIP	Work in progress

Runway **16R/34L is closed due to maintenance or construction work**.



Keyvan Aviation © 2025. All rights reserved.

Weather Satellites/Radars

Observing Weather from Space
Weather satellites provide **large-scale observations of the atmosphere** and help meteorologists monitor global weather systems.
Satellites allow meteorologists to observe:
•Cloud formation and movement
•Storm development
•Jet streams at high altitude
•Tropical cyclones and large weather systems

Weather Radar
Detecting Storms and Precipitation
Weather radar detects **precipitation and storm intensity** by sending radio waves into the atmosphere and measuring the reflected signals.
Radar helps meteorologists identify:
•Thunderstorms
•Heavy rainfall
•Hail
•Convective storm cells

Radar allows meteorologists to monitor rapidly developing storms in real time.
Radar data is also used to issue aviation weather warnings such as:
•**SIGMET**
•**Convective SIGMET (**A weather warning issued for severe thunderstorm activity that may produce hazards such as severe turbulence, hail, strong winds, and heavy precipitation.)
•**Thunderstorm warnings**

Operational Importance for Aviation
Satellite and radar data are essential for **safe flight operations**.
They help pilots and dispatchers:
•Detect hazardous weather along flight routes
•Avoid thunderstorms and convective storms
•Adjust flight routes or altitudes
•Reduce exposure to turbulence and severe weather

Weather Satellites/Radars

Onboard Weather Detection Systems Used by Pilots
Pilots use onboard weather systems to detect and avoid hazardous weather during flight.
Primary onboard equipment includes:

1. Weather Radar
•Detects precipitation and thunderstorm activity ahead of the aircraft
•Helps pilots avoid **heavy rain, hail, and convective storms**

2. Satellite Weather Data (via datalink)
•Provides large-scale weather information such as **cloud systems and storm development**
•Usually received through **satellite communication systems**

3. Lightning Detection Systems
•Detects lightning strikes associated with **thunderstorms**

4. Weather Datalink (e.g., ADS-B / SATCOM)
Provides graphical weather information such as:

- Radar images
- METARs
- TAFs
- SIGMETs

These systems help pilots **identify hazardous weather and plan safe deviations during flight.**

Meteorological Layers in Digital Charts

Modern aviation systems integrate **weather information directly into digital maps**.

Examples of digital aviation tools:

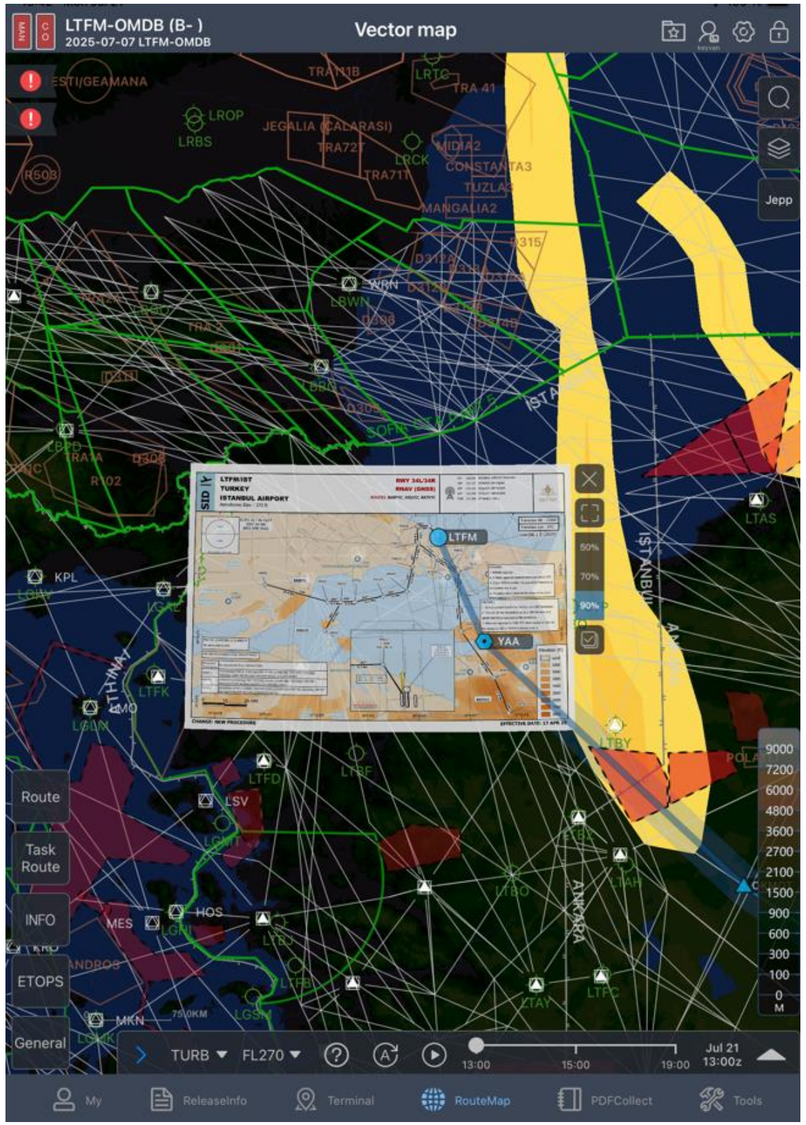
- **Electronic Flight Bags (EFB)**
- **Flight Dispatch Systems**
- **Air Traffic Control Displays**

Provide pilots and dispatchers with **real-time weather visualization** along the route

Pilots can see different weather hazards directly on digital charts:

- **Cloud cover layers** → cloud bases and tops
- **Wind and jet stream data** → strength and direction
- **Thunderstorm areas** → convective cells
- **Turbulence zones** → light, moderate, severe
- **Icing forecasts** → altitude-specific icing conditions
- **SIGMET/AIRMET areas** → hazardous weather warnings

These layers allow pilots to **visualize risks along the planned route**.



EASA Weather Information to Pilots Strategy

In 2018, EASA published the *Weather Information to Pilots Strategy Paper* as part of the All Weather Operations project to improve how weather hazards are communicated to pilots.

The **European Union Aviation Safety Agency (EASA)** developed a strategy to improve how aviation weather information is provided to pilots.

Current Challenges
Traditional aviation weather information is mostly **text-based and fragmented**:
•METAR – current weather observation
•TAF – weather forecast
•SIGMET / AIRMET – hazard warnings
•NOTAM – operational notices

Pilots must **interpret multiple reports separately**, which can increase **workload and the risk of misinterpretation**.

Strategy Goal
Improve **situational awareness and decision-making** by modernizing weather information systems.

The EASA Weather Information to Pilots Strategy is not a regulation but a strategic roadmap guiding the future development of digital and hazard-based weather information systems in aviation.

EASA Weather Information to Pilots Strategy

Key Principles of the EASA Strategy

1. Integrated Weather Information
Combine multiple sources into **one unified weather display**, including:
•METAR / TAF
•Radar and satellite data
•SIGMET / AIRMET warnings

2. Hazard-Based Weather Information
Focus on **operational hazards**, such as:
•Turbulence
•Thunderstorms
•Icing
•Wind shear

3. Digital & Graphical Weather Displays
Future cockpit systems will present weather using:
•**Digital aviation charts**
•**Graphical weather layers**
•**Real-time hazard visualization**

Key Idea:
Weather information should be **easier, faster, and safer for pilots to interpret**.

Most developments are expected to evolve **during the next decade** as part of European aviation modernization programs.

Case Study — Flight: Istanbul (LTFM) → Frankfurt (EDDF)

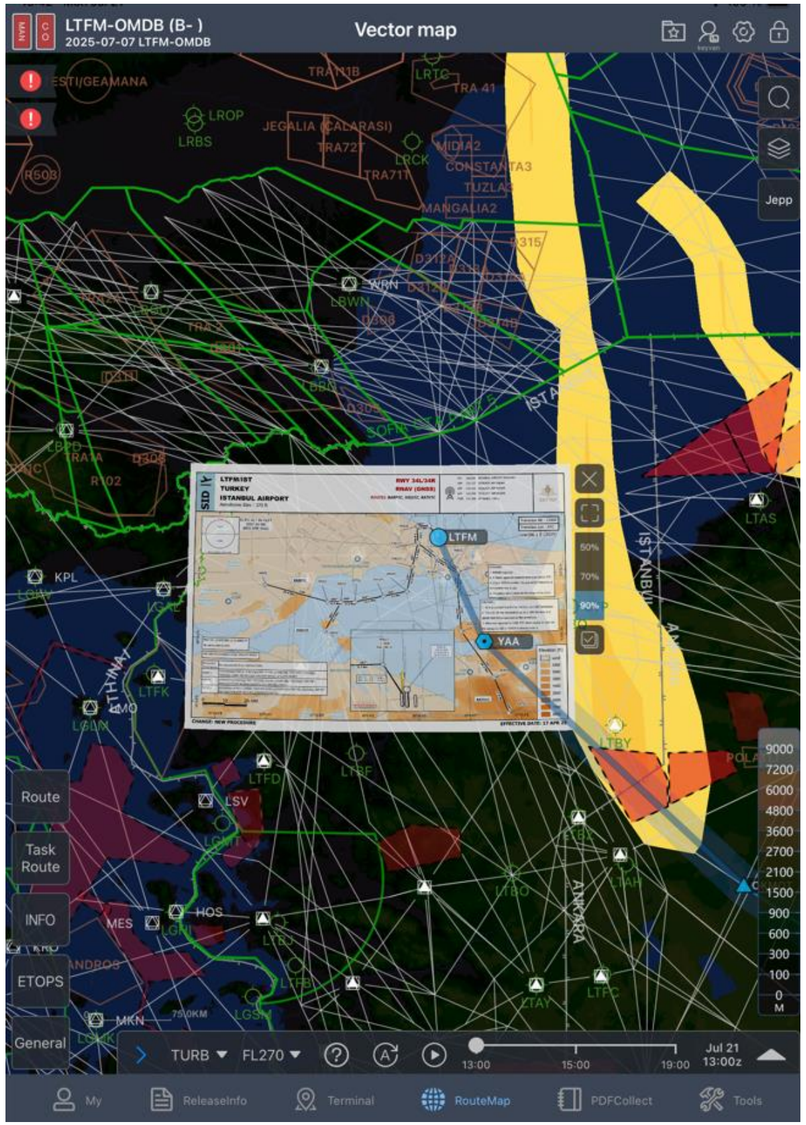
Flight Plan:
•Departure: 12:00 UTC
•Cruise: FL360
•Arrival: 15:30 UTC

Step 1 — Pre-Flight Weather Check (METAR & TAF)
METAR:
```
LTFM 121100Z 03025KT 9999 BKN030 08/03 Q1015
```
TAF:
```
LTFM 121100Z 1212/1312 03020KT BKN030
TEMPO 1212/1215 2000 RA BKN010
```

•Wind 030° at 25 KT → runway selection important
•Broken clouds at 3000 ft → possible IFR departure
•Temporary reduced visibility in forecast (2000 m rain)

Which runway is safest for takeoff given the wind 03025KT?
Runway aligned with wind (e.g., 03R/03L if available).

How does the temporary reduced visibility in TAF affect departure planning?
May require instrument departure (IFR) and extra attention during climb.



Case Study — Flight: Istanbul (LTFM) → Frankfurt (EDDF)

Step 2 — Operational Notices (NOTAM)
NOTAM:

```
RWY 16R/34L CLOSED DUE WIP
Valid 12:00-18:00 UTC
```

•Runway closed → may require alternate runway
•Impacts taxi route and departure clearance

What do pilots need to check before taxiing?
Which runway is available, taxiway changes, ATC instructions.
Could this NOTAM cause delays?
Yes, if traffic is high or alternate runway requires longer taxi.

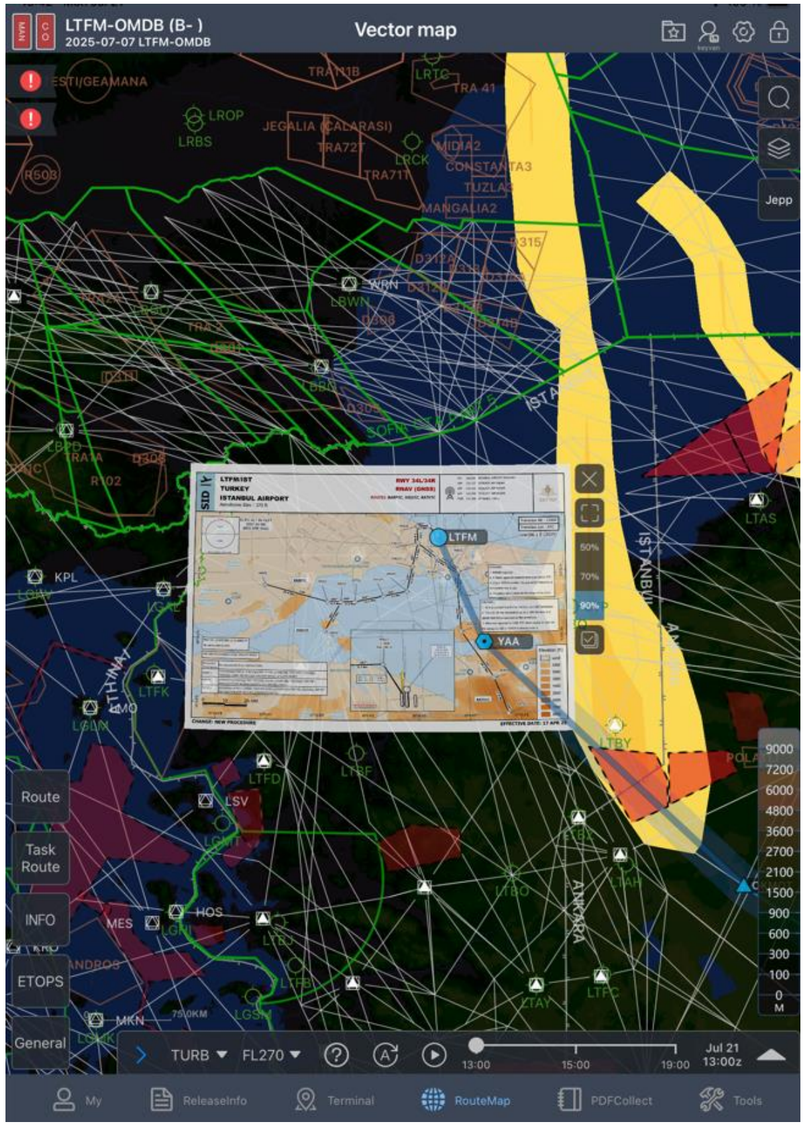
Step 3 — Moderate Weather Warnings (AIRMET)
AIRMET Example:

```
MOD TURB BTN FL080 AND FL180
MOV NE 15KT
```

•Moderate turbulence in climb phase
•Affects aircraft at 8,000–18,000 ft

How should the pilot adjust the climb?
Reduce climb rate, brief passengers, keep seatbelts fastened.

Is this turbulence dangerous for large aircraft?
Usually no; moderate turbulence affects passenger comfort more than safety.



Case Study — Flight: Istanbul (LTFM) → Frankfurt (EDDF)

Step 4 — Severe Weather Warnings (SIGMET)
SIGMET Example:

```
SEV TURB FL300/FL380 OBS AT 1130Z
MOV NE 20KT
```

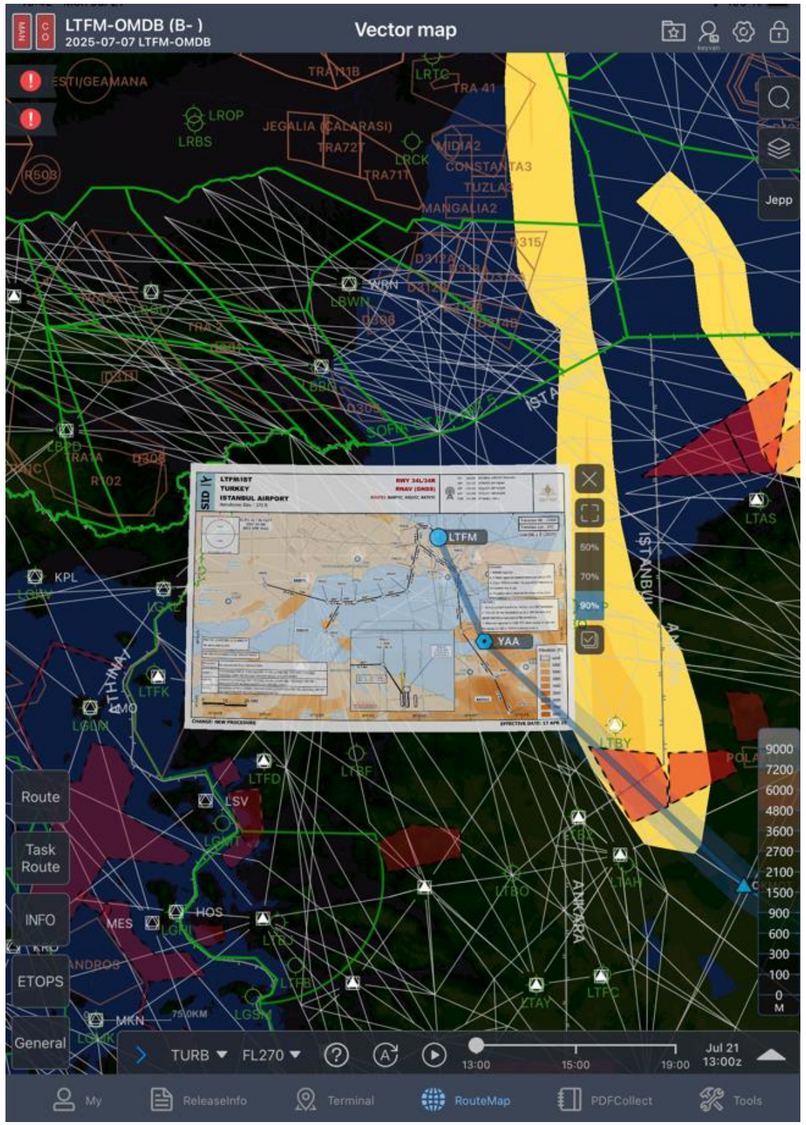
•Severe turbulence at cruise altitude (FL300–FL380)
•Aircraft at FL360 → may need **altitude change** or **reroute**

If your planned cruise is FL360, what should you do?
Change altitude (e.g., FL390 or FL280) or reroute to avoid turbulence.

Why is SIGMET more critical than AIRMET?
SIGMET indicates **severe conditions** affecting all aircraft; AIRMET is moderate.

Step 5 — In-Flight Considerations
Climb Phase:
•Passes through **AIRMET turbulence layer (FL080–FL180)**
•Action: adjust climb rate, seatbelt sign on
Cruise Phase:
•Avoid **SIGMET severe turbulence**
•Monitor **onboard weather radar**
Descent & Approach:
•Check **TAF at Frankfurt**
•Consider NOTAMs affecting arrival/runway

What is the priority when encountering turbulence at cruise?
Safety first → adjust altitude or route, warn passengers.
Which sources should the pilot use to plan descent?
Latest TAF, NOTAMs for destination, AIRMET/SIGMET updates.



Case Study — Flight: Istanbul (LTFM) → Frankfurt (EDDF)

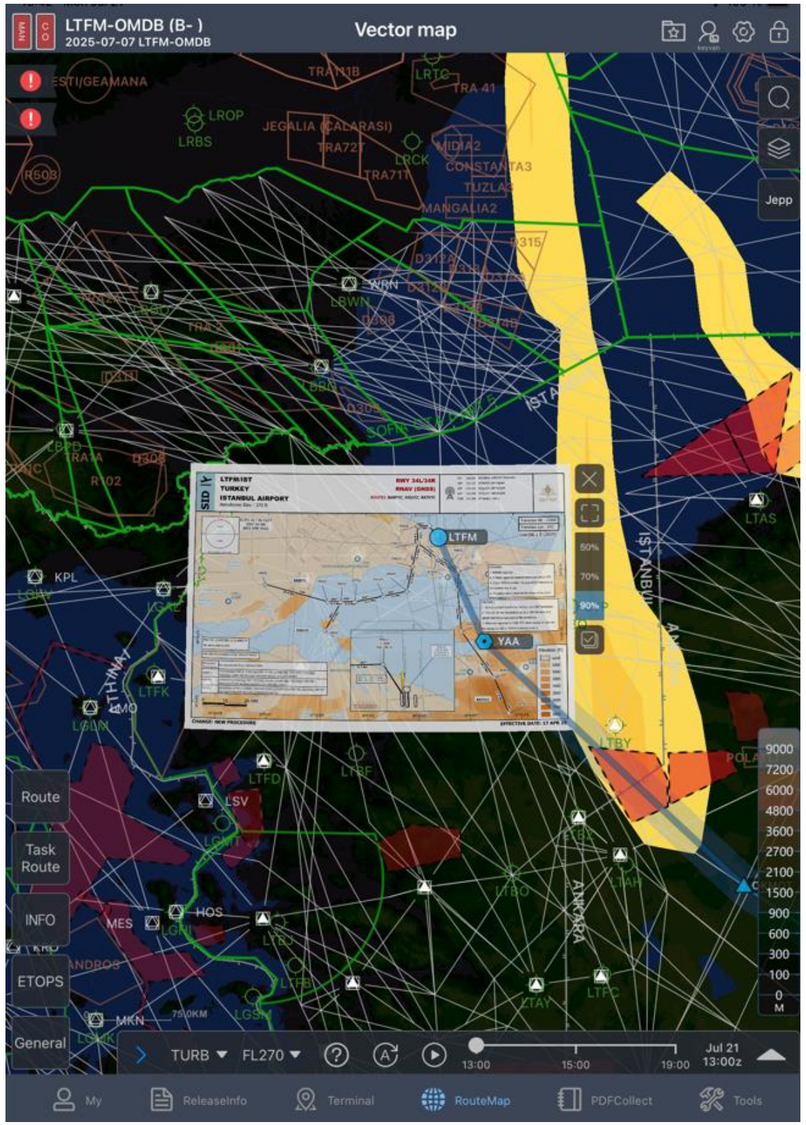
Flow of Weather Information in Flight Planning:

Phase	Source	Purpose
Pre-flight	METAR & TAF	Current & forecast weather
Operational planning	NOTAM	Runway / airport restrictions
Climb & approach	AIRMET	Moderate hazards
Cruise	SIGMET	Severe hazards

Flight Dispatchers (before the flight)
•Analyze METAR, TAF, NOTAM, AIRMET, SIGMET
•Select route and cruise altitude
•Calculate fuel requirements
•Prepare the **flight plan and weather briefing**

Pilots (before and during the flight)
•Review dispatcher briefing
•Monitor updated weather reports
•Adjust altitude or route if conditions change
•Ensure passenger and aircraft safety

Pilots and flight dispatchers must integrate all meteorological and operational information to plan a safe and efficient flight, considering route selection, altitude changes, fuel requirements, and passenger safety.



KEYVAN
AVIATION

THANK YOU

For any questions you may have or clarifications you may require, please kindly contact:

navdb@keyvan.aero

+90(212)2719191

Address:Vadistanbul1BBlockLevel:11Sariyer/Istanbul/Turkey

Website: www.keyvan.aero

/Keyvanaviation